

Project description

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Name: Lone Cirotzki

Table of Contents

Project description	1
<i>Project title</i>	<i>3</i>
<i>Background.....</i>	<i>3</i>
<i>Problem statement</i>	<i>4</i>
<i>Hypothesis</i>	<i>4</i>
<i>Project methodology.....</i>	<i>4</i>
Study design	4
Participants and recruitment.....	4
Intervention	6
Outcomes	7
Statistical power	8
Data analysis.....	8
<i>Ethical considerations</i>	<i>8</i>
Benefits – practical value.....	8
Limitations - risks.....	9
Measures to ensure ethical conduct and publication of the project	9
<i>References</i>	<i>11</i>

Project title

REACT – comparing the health benefits of digital home-based resistance exercise and supervised on-site resistance exercise in cancer patients after treatment.

Background

Cancer is a disease that affects many people and will affect even more in the future. In Norway, more than 330,000 people are currently living with or have previously had cancer (Kreftforeningen). Thanks to ongoing cancer research and advancements in detection and treatment methods, a growing number of people are surviving cancer. On the other hand, cancer survivors often face significant physical, mental, and metabolic health challenges following treatment, including reduced muscle strength, fatigue, decreased quality of life, and increased risk of comorbidities (Bower, 2014; Phillips & Currow, 2010). Therefore, there is a strong demand for sustainable, accessible, and affordable strategies to help reduce and prevent negative health outcomes following cancer treatment.

Exercise has been widely recognized as an effective intervention to counteract these adverse effects of cancer treatment, improving both functional capacity and overall health outcomes in cancer patients (Bower, 2014; Mishra et al., 2012; Speck et al., 2010). Specifically, resistance training is widely recognized as an effective strategy for managing the numerous physical and psychological challenges that often arise after cancer and its treatment (Christensen et al., 2018; Keogh & MacLeod, 2012; Lee, 2022). Resistance training offers significant benefits, including increased muscle mass and improved strength, enhanced functional abilities and quality of life (Keogh & MacLeod, 2012), and positively influencing metabolic and lipid profiles (Christensen et al., 2018). Overall, the benefits of resistance exercise support cancer patients in their rehabilitation and may accelerate their return to everyday life.

However, access to supervised, on-site exercise programs may be limited due to geographic, logistical, or personal barriers. With advancements in digital health technologies, home-based training programs delivered online has emerged as a promising alternative, offering flexibility and increased accessibility. Despite this potential, there is limited evidence directly comparing the effectiveness of digital home-based resistance exercise with traditional, supervised on-site training in cancer survivors. Understanding whether home-based programs can deliver comparable health benefits is crucial to expanding supportive care services and ensuring equitable access to rehabilitation.

Problem statement

The overall research objective is therefore to compare the efficacy of 12 weeks digital, home-based resistance exercise program in cancer patients after treatment on muscular endurance compared to the same program conducted in a studio under guidance of an instructor.

Hypothesis

We predict that participants in both the digital home-based and supervised on-site groups will experience significant gains in muscular endurance after 12 weeks of training, with improvements expected to be similar between the two groups.

Project methodology

This master's thesis is part of a larger ongoing research project titled *Resistance Exercise After Cancer Treatment* (REACT). The project is a collaborative effort between University of Inland Norway (INN) and Innlandet Hospital (SI). It involves a PhD candidate, Anna Solheim, and is led by a project team consisting of Professor/MD Arne Kolstad, Associate Professor Anne Mette Rustaden, and Professor Stian Ellefsen. The main aim of the REACT study is to investigate adherence to exercise in a digitally delivered home-based program compared to supervised, on-site resistance training, as well as to assess a range of functional, mental, and metabolic health outcomes. Round 1 in the data collection was performed during eastern 2024, including 70 participants, while round 2 with the remaining participants will be performed during eastern 2025.

Study design

The project is designed as a randomized controlled trial with assessor blinding (RCT). Participants will be randomly assigned to one of two groups: a 12-week digital, home-based resistance training program performed twice weekly, or the same regimen carried out in a studio setting under the supervision of an instructor. However, since the study aims to determine whether the new intervention – digital home-based resistance exercise – is not significantly less effective than the established on-site program, it also qualifies as a non-inferiority trial.

Participants and recruitment

The study will involve adult cancer survivors (age ≥ 18 years) who have completed hospital-based treatments, including but not limited to chemotherapy, radiotherapy, immunotherapy

and targeted treatment, within the past five years. A minimum life expectancy of 12 months is required. Participants must be capable of providing informed consent. Additionally, they must have access to digital tools at home, including a computer or tablet and an email address, to facilitate login to the digital training platform if assigned to the remote intervention group. Proficiency in the Norwegian language is essential, and individuals should not have engaged in regular resistance exercise (defined as two or more sessions per week) in the last three months. Eligibility is limited to individuals residing within a 0-20km radius of Gjøvik and Lillehammer and who are willing to participate in a resistance training program at either Pusterommet in Gjøvik Hospital or Aktivklinikken in Lillehammer.

Individuals will be excluded if they have medical conditions that contraindicate exercise such as severe cardiac, pulmonary, or other organ-related diseases. Cancer-related complications that limit physical activity, including major surgeries, fractures, or persistent pain, will also serve as exclusion factors. Additional exclusion criteria include a life expectancy of less than 12 months, cognitive impairments that prevent informed consent or program participation, and a residential distance greater than 20 kilometers from Gjøvik or Lillehammer for those considered for on-site training.

Following informed consent, study participants will first be stratified into two groups based on sex and geographic proximity to the training site: (1) women who have completed active treatment and live within approximately 0–20 km of the exercise facility, and (2) men who have completed active treatment and reside within the same distance range. Within each stratum, participants will be allocated in a 1:1 ratio to either a 12-week digital, home-based resistance training program or the same program conducted on-site under supervision.

Randomization will be carried out using block randomization (blocks of 10) and performed by an independent researcher using R software to ensure allocation concealment.

Participant recruitment will be carried out by physicians at the Department of Oncology at Innlandet Hospital (SI), under the supervision of MD/Professor Arne Kolstad. Oncologists will identify patients who meet the eligibility criteria and do not fall under any exclusion parameters, and these individuals will be informed about the study. In addition to clinical recruitment, participants will also be sought through outreach via social media and other public communication platforms such as local papers. Individuals expressing interest will be contacted by the PhD candidate to verify eligibility and provide further details about the study, including the distribution of informed consent forms.

Intervention

Both intervention groups will be prescribed two full-body exercise sessions per week for 12 weeks. The digital group will access a closed group on the www.aktivmotkreft.no digital platform, which provides pre-recorded exercise sessions available anytime. They will be asked to use these pre-recorded sessions and have full flexibility to decide when to perform the exercise sessions. The on-site group participants will follow the same program under the supervision of master's and bachelor's students involved in the project. Exercise sessions will take place at Pusterommet Gjøvik or Aktivklinikken in Lillehammer, with supervision and mentoring from physiotherapists and/or instructors. These sessions will be available on Tuesdays and Thursdays, between 17:30 and 18:30.

The project group, in collaboration with “Aktiv mot kreft”, have developed and pre-recorded the exercise sessions with three different levels to allow for modification and progression. The different levels vary in the number of sets and repetitions: as participants progress, the number of repetitions will decrease while the number of sets increases. Each level will span four weeks. Each session will be approximately 40 minutes long and structured as seen in the table below (Table 1).

Table 1: Exercise Session.

Minutes	What?	How?	Why?
10 min	Warm-up	- Dynamic mobility - Low impact aerobic movements to music with gradually progression to high impact in level 2 and 3. Participants exercising on-site may use ergometercycling and/or treadmill. - Intensity: I-zone 1-3 (BORG RPE intensity scale 10-16, 55-87% of HR_{max})	- Prepare the body for the resistance exercises (e.g. increase the internal temperature and blood flow) - Prepare the mental qualities - Improve physical fitness - Improve cardio metabolic health factors
25 min	Full-body multijoint resistance exercises using body weight, stairs, chair, bottles and resistance band	Intensity: 8-15 repetitions x 2-4 sets, 30-60 second rest between each set - Legs: squat, lunges, step-ups, deadlift, hip thrust - Chest: push-ups, chest press - Back: rowing, back extension - Shoulders: shoulder press, lateral raise - Core: static and dynamic exercises for truncus	- Strengthen muscles and bone mineral density, increase muscle mass and lean body mass - Improve activity of daily living and muscular function - Decrease risk of injury - Improve cardio metabolic health factors
5 min	Cool-down and stretching	- Low impact aerobic movements - Static stretching of large muscle groups – 30 second x 2-3 series - Relaxation	- Maintain or improve mobility - Achieve increased relaxation and reenergizing before returning to everyday life

All participants will receive a resistance band. Adherence will be documented by instructors for on-site/live sessions and via an exercise diary for the digital group. An exercise diary with checkboxes will be used by all participants after each session to record compliance and session experience.

Outcomes

All tests will be conducted at the laboratory at INN, Lillehammer. Standardized protocols will be used for each test. Measurements will be performed at three time points: baseline, within two weeks after completing the 12-week intervention (post-test), and at 12 months follow-up. At each test day, participants will undergo various measurements assessing their functional, mental and metabolic health.

Medical history, cancer diagnosis, and treatment information will be collected via questionnaire and potentially medical records. Participants will also retrospectively register their physical activity level every four weeks during the intervention using a questionnaire accessed via a QR code.

Push-up test

To assess upper body muscular endurance, participants will perform as many correctly executed push-ups as possible, following a standardized protocol. The test can be performed with straight legs, on the knees, or against a wall, depending on the participant's ability. The chosen push-up style, along with the distance between the hands (measured thumb-to-thumb, approximately shoulder-width or up to 5 cm wider), will be recorded. For wall push-ups, the distance from the wall to the tip of the toes is also noted. Participants start with fully extended arms, lower their upper body until there is space for a fist under the chest, and then press back up to full arm extension. Throughout the test, the back must remain straight without breaking at the hips. The total number of continuous repetitions completed without rest will be recorded as the final score. The test will be stopped if the participant can no longer maintain proper technique, defined as two consecutive faulty repetitions.

Keiser Leg Press

To assess lower body muscular endurance, participants will perform repeated leg presses on the Keiser leg press machine at 80% of their previously determined one-repetition maximum (1RM). Before the test, the machine is reset to zero repetitions, and the correct load is calculated as 80% of the participant's 1RM. Participants are instructed to perform each repetition with maximal effort, following a controlled pace set by a metronome at 15 beats per minute, corresponding to approximately 5 seconds per repetition. This pacing ensures that both eccentric and concentric phases are performed steadily within the set time frame. Participants continue the test until they can no longer complete a repetition following the correct pace or technique.

Statistical power

To determine the appropriate sample size, power analyses were conducted using simulations based on three scenarios, where a two-session difference in adherence was considered clinically meaningful. In the primary scenario, equivalence testing assumed a maximum of 24 sessions and ~90% adherence in both groups. Adherence was simulated using a normal distribution ($SD = 4$), and a binomial model was applied with $\alpha = 0.05$. The analysis showed $\geq 90\%$ power to detect equivalence within a 5-percentage-point margin if each group included at least 50 participants. With an expected 15-20% dropout, the target sample size was set at 60 participants per group.

Data analysis

All collected data will be stored securely following GDPR regulations, with personal identifiers removed prior to analysis. Data will be entered into a secure database and cross-checked for accuracy. Descriptive statistics will summarize baseline characteristics. Repeated measures ANOVA will be used to assess changes over time and between groups for primary and secondary outcomes. Where assumptions are not met, non-parametric tests will be applied. Effect size will be reported alongside p-values to provide clinical relevance. Adherence rates will be analyzed descriptively and used as a covariate where relevant. All statistical analysis will be performed using RStudio, with significance set at $p < 0.05$.

Ethical considerations

Benefits – practical value

The REACT project offers several important benefits with both practical and societal value. For participants, engaging in structured resistance exercise can lead to meaningful improvements in muscle strength, functional capacity, metabolic health, and mental well-being, helping cancer survivors regain confidence and improve their quality of life after treatment.

From a broader perspective, the study contributes valuable knowledge on the effectiveness of digital home-based versus supervised on-site exercise programs, which can inform healthcare providers, policymakers, and rehabilitation services. Demonstrating that home-based exercise can deliver comparable health benefits and lead to more accessible, scalable, and cost-effective rehabilitation strategies for cancer survivors, especially for those facing geographic,

logistical, or financial barriers to in-person services. This research thus holds potential to improve the equity and reach of supportive cancer care in the future.

Limitations - risks

While the REACT project is designed to minimize participant burden, some limitations and risks must be acknowledged. Participants may experience temporary discomfort, muscle soreness, or fatigue as a result of the resistance exercise sessions, particularly if they have been previously inactive. These potential drawbacks will be clearly communicated, and all sessions will be adapted to individual capacity and monitored to ensure participant safety.

Additionally, adherence and engagement in the home-based digital program may vary across participants, potentially affecting outcomes and making it harder to compare groups directly. Factors such as access to reliable internet, personal motivation, and self-discipline could influence participation levels in the home-based group, introducing variability. To address this challenge, participants in the digital home-based group will be provided with a training logbook that they will be asked to fill out after each pre-recorded session. In addition, all participants — both in the digital and on-site groups — will receive a short exercise diary with checkboxes to complete after each session. This diary includes questions about compliance (whether they completed all exercises or only part of the session) and their subjective experience of the session. This provides a subjective measure of adherence and perceived experience.

Measures to ensure ethical conduct and publication of the project

This study has been approved by the Regional Committee for Medical Research Ethics in Norway, ensuring that all research activities meet established ethical standards (REK id 553383). All participants will be fully informed about the purpose, procedures, potential risks, and benefits of the study, and will be required to sign a written consent statement prior to participation. Participation is entirely voluntary, and individuals may withdraw from the study at any time without providing a justification or facing any negative consequences. In addition, participants' physical health status will be regularly monitored throughout the intervention period to ensure well-being and address any health concerns that may arise.

All procedures will be conducted in accordance with the ethical principles outlined in the World Medical Association Declaration of Helsinki. Participant data will be handled with

strict confidentiality and stored securely in accordance with the General Data Protection Regulation (GDPR). All personal identifiers will be removed during analysis to ensure anonymity. When publishing results, data will be presented in a way that prevents any possibility of tracing findings back to individual participants. Therefore, the project is considered ethically sound to carry out.

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